INJECTIVITY INTO A PRODUCT OF LOCALISATIONS IS EQUIVALENT TO INJECTIVITY OF THE MAP MULTIPLYING BY THE ELEMENTS

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Allowed standard prerequisites (editorial).

- Vanishing in a localisation as annihilation by a power. Used at the start of the counted proof (lines 4635-4637). Rewrites injectivity of the map into the product of localisations as a condition on elements killed by powers of the f_i.
- Induction on the sum of the exponents. Used in the converse direction (lines 4643-4647). Reduces the general annihilation data to the case where all exponents are one.

Lemma 0.1. *Let R be a ring. Let $f_1, \ldots, f_n \in R$. Let M be an R-module. Then $M \to \bigoplus M_{f_i}$ is injective if and only if*

$$M \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} M, \quad m \longmapsto (f_1 m, \ldots, f_n m)$$

is injective.

Proof. The map $M \to \bigoplus M_{f_i}$ is injective if and only if for all $m \in M$ and $e_1, \ldots, e_n \geq 1$ such that $f_i^{e_i} m = 0$, $i = 1, \ldots, n$ we have $m = 0$. This clearly implies the displayed map is injective. Conversely, suppose the displayed map is injective and $m \in M$ and $e_1, \ldots, e_n \geq 1$ are such that $f_i^{e_i} m = 0$, $i = 1, \ldots, n$. If

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$e_i = 1$ for all i, then we immediately conclude that $m = 0$ from the injectivity of the displayed map. Next, we prove this holds for any such data by induction on $e = \sum e_i$. The base case is $e = n$, and we have just dealt with this. If some $e_i > 1$, then set $m' = f_i m$. By induction we see that $m' = 0$. Hence we see that $f_i m = 0$, i.e., we may take $e_i = 1$ which decreases e and we win. $\square$